

Research Article

Strava Metro mobility data provides accurate estimates of recreation use in Urban-Proximate Protected Areas

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ABSTRACT

This study evaluates the efficacy of Strava Metro data for estimating recreational trail use in protected areas, which targets a critical need to understand patterns of human movement given the increasing trends in visitation. We compare Strava Metro data, encompassing aggregated pedestrian and cycling activity, with calibrated TRAFx trail counter data across 13 trails within three protected areas in the greater Los Angeles region over a one-year period. Along with the Strava Metro data, we collected daily weather covariates to analyze the efficacy and utility of Strava Metro for predicting recreational trail use. Our findings demonstrate the potential of Strava Metro to provide robust estimates of recreational trail use, particularly for cycling use, while also acknowledging inherent limitations concerning representativeness and potential socio-demographic biases. This research contributes to the development of effective monitoring strategies using secondary data sources to understand human mobility patterns in protected area settings that inform park management decisions and support long-term conservation objectives.

Management Implications

Monitoring recreation use in protected area settings is fundamental to sustainable planning and management, however traditional monitoring techniques can be cost and resource intensive leaving managers with limited or no information. This research demonstrates that *Strava Metro* data, when combined with weather and temporal covariates, offers protected area managers a cost-effective tool for estimating daily recreation use at fine spatial and temporal scales. We show that models can produce highly accurate daily use predictions, particularly for cycling-focused trails, enabling managers to better understand weekly and seasonal use patterns, inform operational decisions like trail maintenance schedules, and strategically allocate resources for visitor safety during peak-use periods.

However, we show that *Strava Metro* data is not demographically representative, with significant bias toward male cyclists aged 35–54 while under-representing pedestrians, females, and older adults. Therefore, while managers can leverage *Strava Metro* for quantifying use, it cannot replace traditional survey methods for understanding visitor characteristics. Instead, we recommend an

integrated approach, combining the continuous monitoring of *Strava Metro* with periodic visitor-intercept surveys to ensure management decisions support both conservation goals and high-quality recreation opportunities for all visitors.

1. Introduction

Protected areas worldwide are major focal points for nature-based recreation while also serving as critical habitat for biodiversity conservation. Balmford et al. (2015) estimated that at the global scale terrestrial protected areas receive roughly 8 billion visits per year, and recent trends suggest increasing visitation to protected areas in the United States. According to the National Park Service (NPS), visitation to national parks increased by 19 % from 2014 to 2019. Since the COVID-19 pandemic, National Park visitation has steadily increased back toward pre-pandemic levels, with a 4.3 % increase from 2022 to 2023 (Ziesler & Spalding, 2024). Similarly, the U.S. Forest Service reported a 5 %

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increase in National Forest System visitation between 2014 and 2018 (USDA, 2023), and a 10 % increase between 2019 and 2020 (Avitt, 2021). Although the cause of this increasing trend in protected area visitation is not well understood, it is often attributed to several factors such as increasing awareness of the benefits of outdoor recreation, growing interest in ecotourism, and the influence of social media (Wichman, 2024; Winter et al., 2019). Nevertheless, these increases in visitation present challenges for visitor use management, where increasing visitation can influence perceptions of crowding and safety (Manning & Valliere, 2001; McCool & Christensen, 1996, pp. 67–83; Nettles et al., 2021) as well as increase the potential for ecological disturbance associated with recreation (Buckley, 2013; Leung & Marion, 2000) which in turn can influence the quality of the visitor experience (Farrell et al., 2001; Lynn & Brown, 2003).

Managing recreation use in a manner that is compatible with the conservation goals of protected areas is a critical challenge and requires an understanding of the interactions and responses between recreation visitors and the environment. The functional responses of biotic (e.g., soils and vegetation) and abiotic (e.g. water) resources to recreation are generally well understood (Cole, 1995, 2004; Hadwen et al., 2008; Monz et al., 2013), but recent inquiry has highlighted the importance of understanding human presence which can be a key driver of spatial and temporal patterns of wildlife movement (Benson et al., 2021; Larson et al., 2019; Salvatori et al., 2023; Tucker et al., 2023; Zeller et al., 2024). Consequently, there is a growing need for effective methodologies to provide reliable estimates of recreation use and human presence, or dynamic human footprint, particularly in protected area settings, at fine spatial and temporal scales (Ellis-Soto et al., 2023).

Obtaining reliable estimates of recreation use and human presence in protected areas is challenging, as there are no uniform standards for reporting visitation among land managers in the United States, nor a consistent definition of a recreation visit (Hanson et al., 2025). As a result, research has increasingly turned to novel data sources (e.g., social media and visitor generated data) to estimate recreation use. However, these data sources introduce new challenges for the generalizability and representativeness of the data, but also provide opportunities to model and estimate human presence in protected areas.

1.1. Traditional and emerging recreation use monitoring techniques

Traditionally, recreation use-estimation has employed automatic trail-counter devices (e.g., *TRAFx*, and *EcoCounter*) to monitor non-motorized recreation. These devices can be labor intensive to install and maintain, and often require direct observation to calibrate the instrument to provide reliable estimates of recreation use (Pettebone et al., 2010). Given the many competing priorities for protected area management and limited resources, these traditional monitoring techniques are often installed for a short period of time or at key locations. Additionally, in many protected-area contexts, the boundaries of the protected area are porous, with several primary and secondary entrances which increases the challenge in designing a trail-counter monitoring approach capable of providing accurate estimates of recreation use. However, recent research has highlighted the utility of social media and volunteered geographic information (VGI) data for estimating recreation use, which can provide a cost-effective and scalable alternative to traditional approaches.

A growing corpus of research, motivated by the growing availability (Teles da Mota & Pickering, 2020) and influence (Wichman, 2024) of social media and other visitor generated data, has demonstrated many techniques that incorporate novel visitor-generated data to estimate visitor use (Wilkins et al., 2024). These studies have relied on a wide variety of digital data streams, such as social media apps like Twitter, Flickr, and Instagram (Fisher et al., 2018; Tenkanen et al., 2017; White et al., 2023b; Wilkins et al., 2020, 2021b; Wood et al., 2013, 2020), fitness tracking apps (Leggett et al., 2017; Norman & Pickering, 2019; Reinhardt et al., 2024), mobile device data (Creany et al., 2021; Monz

et al., 2019; N.P.S., 2020; Whitney et al., 2023; Sergeyev et al., 2025; Rice et al., 2025), and other forms of digital trace data (Rice & Pan, 2021) that can be used to estimate visitor use. Many of these studies have used data sources that provide a proxy for recreation use, such as photo-user-days with Twitter and Flickr, which require localized calibration to scale these global datasets in order to provide reliable estimates of recreation visitation (Tenkanen et al., 2017; Wilkins et al., 2021b; Wood et al., 2013, 2020). Additionally, because the active use of any particular social media platform can fall in and out of vogue over time, the best techniques often use an ensemble of social media data sources and weather covariates (Tenkanen et al., 2017; Winder et al., 2025; Wood et al., 2020), yet they still require careful interpretation as these data sources are not always representative of the entire population of visitors (Sessions et al., 2016).

A separate line of research has relied on a more direct method to estimate visitor use or understand visitor behavior from Volunteered Geographic Information (VGI) that draw from various sources including Wikiloc

(Campelo & Nogueira Mendes, 2016; Norman & Pickering, 2017), GPSies (Campelo & Nogueira Mendes, 2016), MapMyFitness (Norman & Pickering, 2017), *Strava* (Campelo & Nogueira Mendes, 2016; Reinhardt et al., 2024; Rice et al., 2019), as well as custom smartphone-based apps (Korpilo et al., 2017; Muñoz et al., 2019). While these VGI data sources are subject to the same fluctuations in popularity of use as photo sharing apps, *Strava* stands out as an enduring and dominant repository for activity tracking with 135 million registered users and 6 billion recorded activities (Strava, 2025).

In recent years, an increasing number of studies have used *Strava* data to inform urban planning and active transportation (Boss et al., 2018; Hochmair et al., 2019; Dadashova et al., 2020; Jestico et al., 2016;

Nelson et al., 2021), understand recreation behavior and spatial patterns (Minehart et al., 2024; Rice et al., 2019), assess anthropogenic disturbance on wildlife (Corradini et al., 2021), and as a technique to estimate recreation use in protected areas (Reinhardt et al., 2024; Venter et al., 2023). Reinhardt et al. (2024) demonstrated that data from *Strava Metro*, a service that aggregates and anonymizes *Strava* user data, had a strong relationship with trail counter data and accurately represented variation in the spatial and temporal patterns of visitor use. However, the generalizability and bias of these fitness tracking apps, which are often not representative of the population of park and protected area (PPA) visitors, has often been cited as a limitation and concern. Venter et al. (2023) found *Strava* data had diminishing relationship with recreation use from monthly to weekly and daily scales, but noted the data exhibited significant biases along gender, age, education, and socio-economic status. Acknowledging that no source of data is free of bias, these limitations of social media data are important considerations when using these data to inform protected area planning and management.

1.2. Influence of climate and weather on recreation use

Recent research motivated by understanding the effect of changing climate on recreation use has evaluated the influence of current weather and long-term climate patterns on visitor behavior and trends in recreation use (Fischelli et al., 2015; Wilkins et al., 2025). Studies spanning diverse settings, including U.S. national parks, reveal a strong association between visitation and temperature, where visitation generally increases with warmer average temperatures up to a comfort threshold, but declines sharply once temperatures exceed very warm levels. Fischelli et al. (2015) found that visitation to U.S. National Parks decreases at mean monthly temperatures above 25 °C. Additionally, climate projections with rising temperatures are expected to contribute to an expansion of the peak visitation seasons in some settings (Wilkins et al., 2021a), and influence decision making about the activity-type (White et al., 2023a; White et al., 2023b), and timing and location of visits

(Wilkins et al., 2021b). Beyond temperature, humidity is strongly

connected with outdoor thermal comfort, affecting thermal safety and increasing heat-related illness risks (Buttke et al., 2023; Craig, 2024), while precipitation and wind generally have limited or variable impacts on visitation (Wilkins & Horne, 2024). Ultimately, weather effects often interact with non-climatic factors such as holidays or weekends, which frequently have greater influence on visitation than weather variables alone (Jaung & Carrasco, 2021).

The primary aim of this study is to provide a robust assessment of the efficacy of *Strava Metro* data to predict daily trail use with a model-based approach. A secondary objective is to evaluate the generalizability and representativeness of *Strava Metro* data and quantify the extent of the bias among *Strava* users to help inform future efforts using *Strava Metro* data in peri-urban protected area settings.

2. Methods

2.1. Study area

This study is focused on 13 multi-use trails within three peri-urban protected areas in Orange County, California, U.S.A., located approximately 50 km south of Los Angeles, CA (Fig. 1). The two coastal protected areas (i.e. Laguna Coast Wilderness and Aliso and Wood Canyons Wilderness) are popular recreation destinations with coastal vistas and canyon terrain and provide mixed-use/multi-use trails for hiking, cycling, and equestrian use. Similarly, Santiago Oaks Regional Park located in the inland chaparral landscape provides multi-use trails for

hiking, cycling, and equestrian use, but features trails designed specifically for mountain biking. Across the three parks, the trails in this analysis are primarily single-track (i.e., 50 cm-1m wide) apart from the Sage Ridge trail which is a wider (2m) trail.

Trail use was monitored continuously using *TRAFx* infrared counters between May 2021 to April 2022, which were calibrated using best practices (Pettebone et al., 2010). The *TRAFx* data were assessed for erroneous and outlying points (i.e. $> |3 \text{ sd}|$) against values for the same hour of day and day of week. Anomalous or outlying points were interpolated using a univariate derivative spline via Scipy (Virtanen et al., 2020). Throughout the course of the data collection period, volunteers conducted 259 calibrations of the *TRAFx* counters which provided a robust assessment of measurement error and were used to calculate a correction coefficient for each counter. These observations provided additional information regarding the proportion of cyclists and pedestrians and the direction of travel (i.e., uphill vs. downhill) as trail users passed the counter. These direct observations of the type and direction of use were aggregated and averaged to classify the prevailing activity-type and direction of use for each trail.

2.2. Data collection and preparation

Next, using the *Strava Metro* web-interface, we queried daily *Strava* aggregated data from pedestrians and cyclists for each trail throughout the temporal extent of the *TRAFx* data. Although *Strava Metro* data were available at higher temporal resolutions (i.e. hourly), we elected to use



Fig. 1. Reference map providing an overview of the three protected areas within Orange County, California. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

daily counts for this analysis spanning a calendar year because of how *Strava Metro* processes recorded activities. *Strava Metro* rounds the counts of recorded activities on a trail segment to the nearest 5 as a measure to maintain the privacy of *Strava* user data. Consequently, data for trails with low use (i.e., less than 3 activities per temporal unit) is more likely to be limited and previous studies have shown that the precision of *Strava Metro* data decreases at finer temporal scales (Venter et al., 2023). We did not perform any transformations or processing to the *Strava Metro* data, aside from summing the daily totals of pedestrians and cyclists to create a *Strava Total* variable. *Strava Metro* data includes descriptive metadata related to the recorded activity and the users, such as the direction of travel along the trail, demographic information about the users, and behavioral data (e.g. velocity). We evaluated the dataset to assess the normality of the *Strava Metro* data and calculated a Spearman's ρ partial correlation coefficient to determine the strength of the relationship between the *Strava* and *TRAFx* data. We calculated Spearman's ρ for each trail segment, and for each activity-type (i.e. pedestrians, cyclists, mixed) to gain a better understanding of what activity types were most well represented with *Strava Metro*. Finally, we paired the *TRAFx* and *Strava Metro* datasets based on the date and trail segment to create a single dataset for analysis. The final dataset included 3736 daily observations across 13 trails from May 2021 to April 2022.

2.3. Model construction

After merging the *Strava Metro* and *TRAFx* datasets, we added daily weather covariates via the *meteostat*

API (Lamprecht, 2024) which can influence the amount of recreation use. We gathered daily weather records based on the park location that included the daily maximum, minimum, and average temperature, precipitation, and wind speed. Additionally, we added temporal covariates including the day of the week, the length of daylight, and the day of the year to the dataset to account for weekly and seasonal trends in recreation use. We elected to use a CatBoost (Dorogush et al., 2018) gradient boosted regression tree model to predict daily trail use due to its flexibility in handling and regularizing categorical and continuous predictors, as well as its ability to handle non-linear relationships between the predictors and the response variable. Next, we conducted an optimization of the model hyperparameters using Optuna (Akiba et al., 2019) to minimize the root mean squared error (RMSE) of the model predictions. We used a randomly drawn subset of 70 % of the observations for model training and fitting, and evaluated model performance on the remaining 30 % of the dataset.

2.4. Demographic data

To evaluate potential biases of *Strava Metro* data, we used demographic data from a random visitor-intercept survey, collected at each trail in 2021 and 2022 as part of the larger study to evaluate the effects of trail management strategies. In total, there were 1140 survey responses across the study area trails, and included data on age, gender, and activity type. We aggregated and summarized the demographic metadata from *Strava Metro* and adapted the survey data variables for age, gender, and activity type by expanding or condensing the categorical variables to align with the *Strava Metro* data. We used a chi-square test of homogeneity to determine if there were any significant differences in proportions between the two data sources.

3. Results

Although the size of daily *Strava Metro* observations was robust, the data were not evenly distributed across the trails and were skewed towards trails whose primary activity-type was cycling (61 %), followed by and mixed-use (27 %), and pedestrian (13 %). We found that the Spearman correlations indicated a strong, positive relationship between daily *TRAFx* counts and *Strava Metro*, $r(3,736) = .712, p < .001$. Partial

correlation analyses, controlling for trail use type, further demonstrated strong relationships between *TRAFx* counts and *Strava Metro* for trails with mixed-use ($r(994) = .804, p < .001$) and trails frequented by cyclists ($r(2,248) = .804, p < .001$). On trails used predominately by pedestrians, the relationship between *Strava Metro* and the *TRAFx* daily counts was considerably weaker ($r(494) = .378, p < .001$).

3.1. Daily visitation model

The daily visitation model demonstrated strong predictive performance (Fig. 2) using *Strava Metro* data and weather covariates to predict daily trail use in the study area PPAs. Model predictions on the training dataset were within ± 12.6 of the observed trail counts (MAE = 12.605) as measured by the *TRAFx* counter, and the model accounted for 97.1 % of the variance in the counts ($R^2 = .971$). Model performance on the validation dataset was also robust, with prediction error of ± 28.2 trail users (MAE = 28.194), and showed strong performance generalizing to new data, explaining 86.0 % of the variance in the *TRAFx* data ($R^2 = .860$). We found the model performed better on weekdays ($R^2 = .860$, MAE = 25.581), compared to weekends ($R^2 = .776$, MAE = 41.845), however the ratio of weekdays and weekends was imbalanced in both training and validation datasets, with a approximately five times as many weekdays ($n = 940$) in the validation dataset than weekends ($n = 181$). The model residuals were normally distributed, notwithstanding a few extreme values in the test dataset, and model predictions exhibited a strong linear relationship with the observed values.

Using the validation dataset, we assessed model performance on a trail-by-trail basis (Table 1), and found the model performed best on trails where the primary use-type was cyclist (e.g. Laguna Ridge) or mixed (e.g. Grasshopper), but performance diminished on trails predominately used by pedestrians (e.g. Pony Trail) as well as those with fewer *Strava Metro* records.

Next, to visually inspect the quality of the model predictions, we plotted the model predictions against the *TRAFx* counts from May 2021 to April 2022 for two trails: Cholla (Fig. 3), a trail used predominately by cyclists, and Grasshopper (Fig. 4), a mixed-used trail. The predicted values on both trails appear to reflect the seasonal and weekly variations and patterns in trail use, with lower use in the late summer months and on weekdays, and higher use on weekends. The fit of the predictions on the Cholla trail is very close throughout the model year, while the predictions for the Grasshopper trail tend to under-estimate the weekend spikes in use from October to March relative to the observed *TRAFx* counts.

Finally, we inspected the feature importance of the Catboost model to understand the contribution of the variables in the model to the predictions of daily trail use (Fig. 5). The most important variable in the model was the total number of *Strava Metro* records, followed by the day of the week, the length of day, and the prevailing direction of use. Interestingly, the direction of trail use (uphill/downhill) appeared to be more important in the model than the trail itself, which suggests that the model is capturing the trends of trail use behavior. Additionally, we found that temporal features such as the day of week and length of daylight were marginally important in the model, but weather-related features (e.g. minimum/maximum temperature) had minimal influence on model tree-splitting relative to the total number of *Strava Metro* counts for a trail.

3.2. Demographic comparison of strava metro and survey samples

We compared the demographic characteristics of *Strava Metro* data against a random visitor-intercept survey sample collected during the summer of 2021 to evaluate potential biases in the *Strava Metro* data using a chi-square test of homogeneity. We found that the demographics of *Strava Metro* users differed significantly from the survey sample, with higher proportions of males and cyclists, and a greater proportion of adults age 35–54 than the survey sample (Table 2). In both datasets the

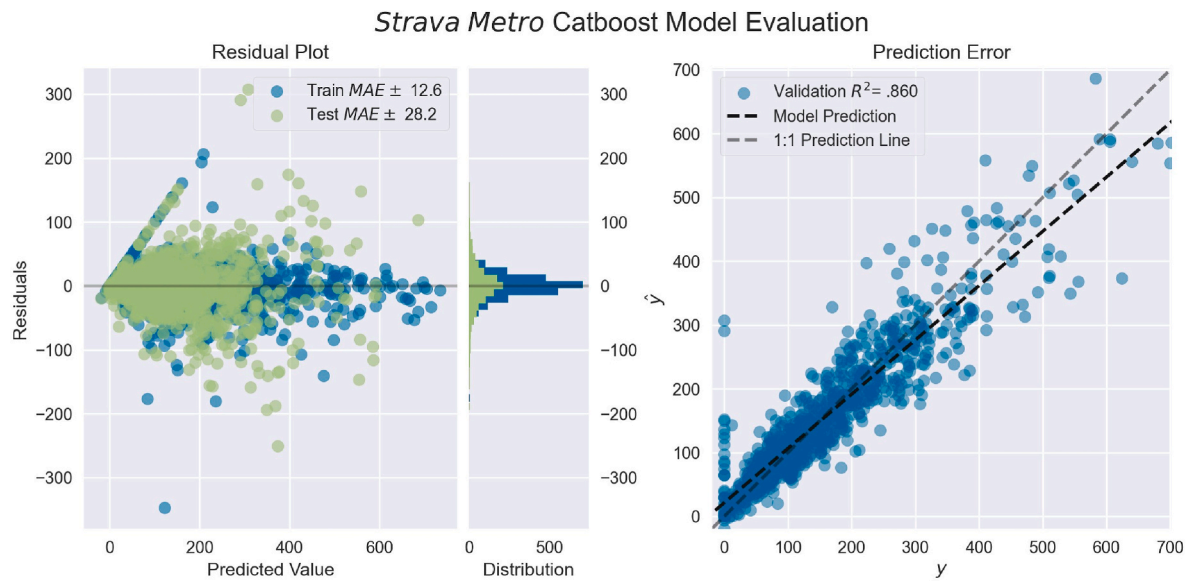


Fig. 2. Catboost regression model diagnostic plots illustrating the model residual error on the training and validation dataset, and the model prediction error on the validation dataset.

Table 1

Comparison of model performance by trail in study area PPAs evaluated using root mean squared error (RMSE), mean absolute error (MAE), and R2 statistics. Trails primarily used by cyclists and where use was mixed had the best performance, while model performance was lower on trails predominately used by pedestrians.

Trail	Primary Use-Type	RMSE	MAE	R ²
Laguna Ridge	Cyclist	16.482	12.678	.914
Cactus Canyon	Cyclist	36.512	26.550	.887
Grasshopper	Mixed	28.224	19.104	.868
Cholla	Cyclist	56.432	39.934	.859
Lizard	Mixed	48.784	35.800	.855
Yucca Ridge	Cyclist	27.896	19.370	.853
Lynx	Cyclist	27.935	20.028	.809
Peralta Hills	Pedestrian	35.988	24.918	.795
Rock It	Cyclist	57.843	39.682	.790
Chutes Ridgeline	Cyclist	45.112	32.406	.787
Pony Trail	Pedestrian	39.645	29.296	.745
Old Emerald	Cyclist	37.927	27.177	.711
Sage Ridge	Mixed	55.269	29.087	.613

majority of trail users were male, though the *Strava Metro* data had a much higher proportion (93.5 %) compared to the survey sample (73.2 %). Additionally, while the survey dataset was nearly evenly split between pedestrians and cyclists (50.8 % and 49.2 % respectively), the *Strava Metro* data was substantially overrepresented by cyclists (93.6 %) compared to pedestrians (6.4 %). Finally, we found that the proportions of trail users age 18–34 were quite similar between the *Strava Metro* (24.4 %) and survey sample (23.4 %), but trail users age 35–54 were overrepresented by *Strava Metro* (62.9 % vs 43.9 %), while trail users age 55–64 and 65+ were underrepresented relative to the survey sample (12.4 % vs 21.1 % and .3 % vs 8.1 % respectively).

4. Discussion

4.1. Using *Strava Metro* data to estimate trail use

This study evaluated the efficacy of using *Strava Metro* data to produce daily estimates of trail use in protected areas. Using visitor-generated data like *Strava Metro* to monitor recreation use in protected areas can offer a more cost-effective approach than traditional monitoring techniques as well as access to a longitudinal dataset to track

trends in use across time. Our model, composed of *Strava Metro* data and weather covariates, provided reasonably accurate estimates (MAE $\pm$ 28.2 trail users, $R^2 = .860$) of daily trail use, which we validated against observed trail counts collected via an automated infrared trail counter (TRAFx). According to the normalized feature importances of the model, *Strava Metro* data was the dominant feature in the dataset for model fitting, followed by day of the week, length of day, and the prevailing direction of trail use (uphill/downhill). We found *Strava Metro* data to have a strong linear relationship with observed trail use, but the strength of this relationship was strongest for trails known to be primarily used by cyclists and where use was mixed with cyclists and pedestrians. Lastly, we evaluated the representativeness of *Strava Metro* data by comparing it against a stratified random visitor-intercept survey sample collected in 2021 and 2022 on the study area trails, and found significant differences in gender, activity-type, and age, with *Strava Metro* data being more likely to be over-represented by males and cyclists, and a greater proportion of adults age 35–54 than the survey sample.

4.2. Model performance

Our results are consistent with a number of studies in different contexts (Reinhardt et al., 2024; Venter et al., 2023) that found *Strava Metro* data to have a strong linear relationship with observed trail use, but we demonstrated these relationships can be improved with localized parameterization to provide accurate estimates of trail use at fine temporal scales. As a result of the ease of use and popularity of *Strava* among trail users, the coverage of *Strava Metro* data is widespread, from the Northeast United States (i.e. Vermont) (Reinhardt et al., 2024), to Northern Europe (i.e. Norway) (Venter et al., 2023), to Southern California. However, the utility of *Strava Metro* data is setting and context dependent, as it relies on visitors recording their activity. Given the multitude of fitness tracking apps, some studies report less frequent use of *Strava* (Schwietering et al., 2024) compared to other services, and in some settings visitors may deliberately choose not to record their activity. Taken together, these studies suggest that *Strava Metro* data can provide a useful tool for estimating human presence and movement behavior and, in the absence of other data, can help inform planning and decision-making regarding patterns of recreation use.

Similar to Venter et al. (2023), we found *Strava Metro* data was highly correlated with trail counts collected with automated infrared trail counters and found similar strength of relationships with the

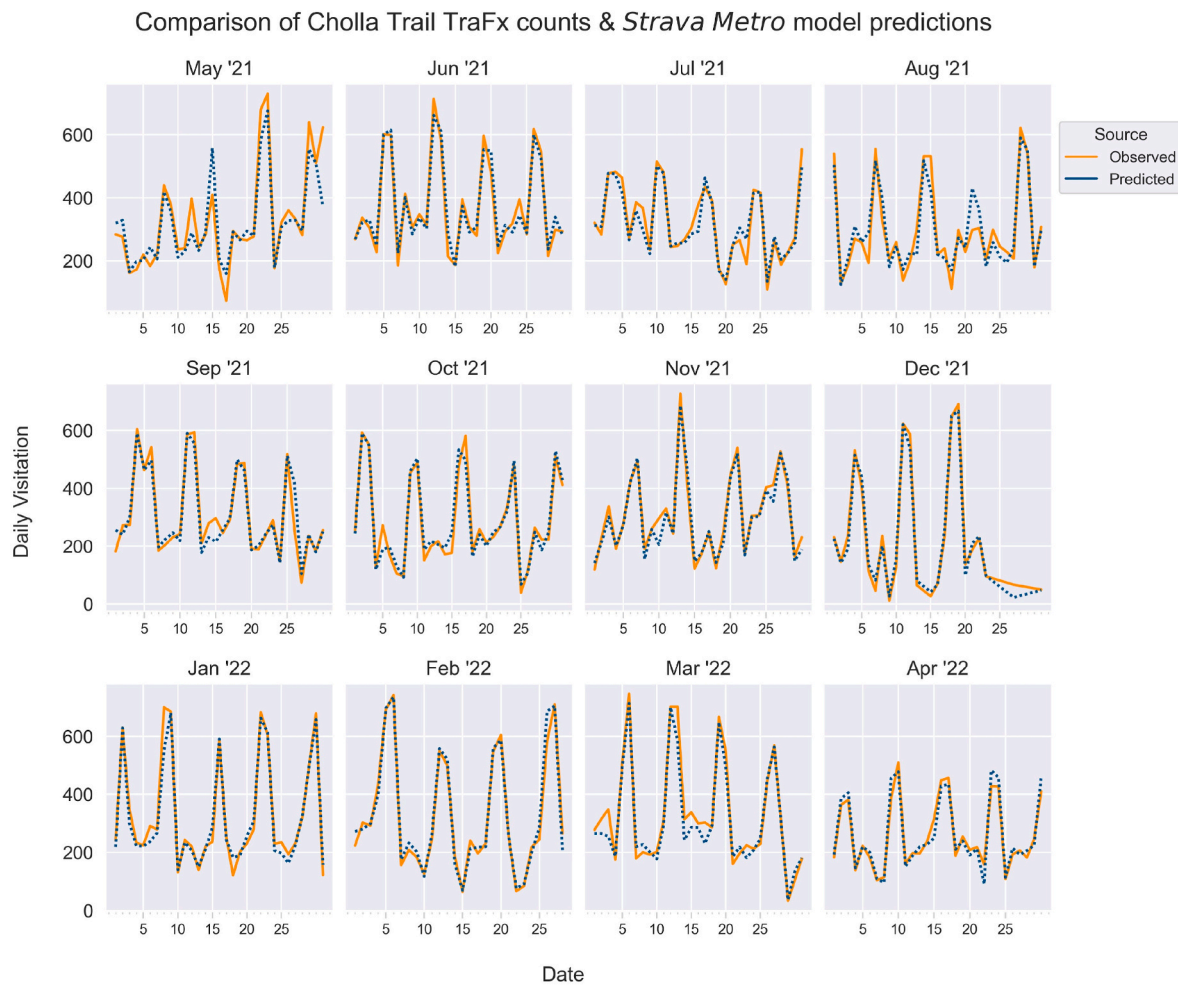


Fig. 3. Daily trail use predictions (blue dotted line) for Cholla trail plotted against the TRAFx data (orange solid line) from May 2021 to April 2022. Generally, the peaks of the daily visitation lines correspond to weekend days, while the troughs correspond to weekdays. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

temporal and spatial variation of trail use. However, comparisons of model performance to Venter and Reinhardt are limited due to the different temporal scales of the data (monthly vs daily), differing approaches for querying data from *Strava Metro*, and different modelling approaches. Our modelling approach differed from Venter and Reinhardt with the inclusion of weather covariates and contextual features which improved the accuracy of estimates ($R^2 = .860$) of daily trail use across the study area. Despite the difference in approaches, these studies demonstrate that *Strava Metro* data can be a useful tool for estimating trail use in a broad range of contexts and at various temporal scales.

Nevertheless, *Strava Metro* is one among many sources of visitor generated data and social media platforms like Twitter, Instagram, and Flickr (Fisher et al., 2018; Tenkanen et al., 2017; White et al., 2023b; Wilkins et al., 2020; Wilkins et al., 2021b; Wood et al., 2013, 2020) and mobile device data (Creany et al., 2021; Winder et al., 2025; Monz et al., 2019; N.P.S., 2020; Whitney et al., 2023; Sergeyev et al., 2025) that have been used to produce estimates of trail use in protected areas.

Studies using these data sources vary in quality of the information and the ability to predict on-site recreation use. For example, Tenkanen et al. (2017) found strong relationships with official visitation statistics in Finland and South Africa ($R^2 = .75-.84$) by combining Instagram, Flickr, and Twitter data at monthly scales. Similarly, Winder et al. (2025) using mobile device datasets and Wood et al. (2020) using Flickr, Twitter and Instagram data reported very high R^2 values (.92 and .91, respectively) for predicting monthly visitation. Our model compares favorably with these studies in terms of model performance, but with the notable difference that *Strava Metro* data provide a finer, daily resolution of trail use. Finer scale temporal resolution of recreation use offers several key advantages, including the ability to better understand the daily and weekly variations in patterns of use, the influence of climate and weather on recreation use, as well as the potential for integration with other data streams to understand interactions and effects on wildlife presence and movement behavior.

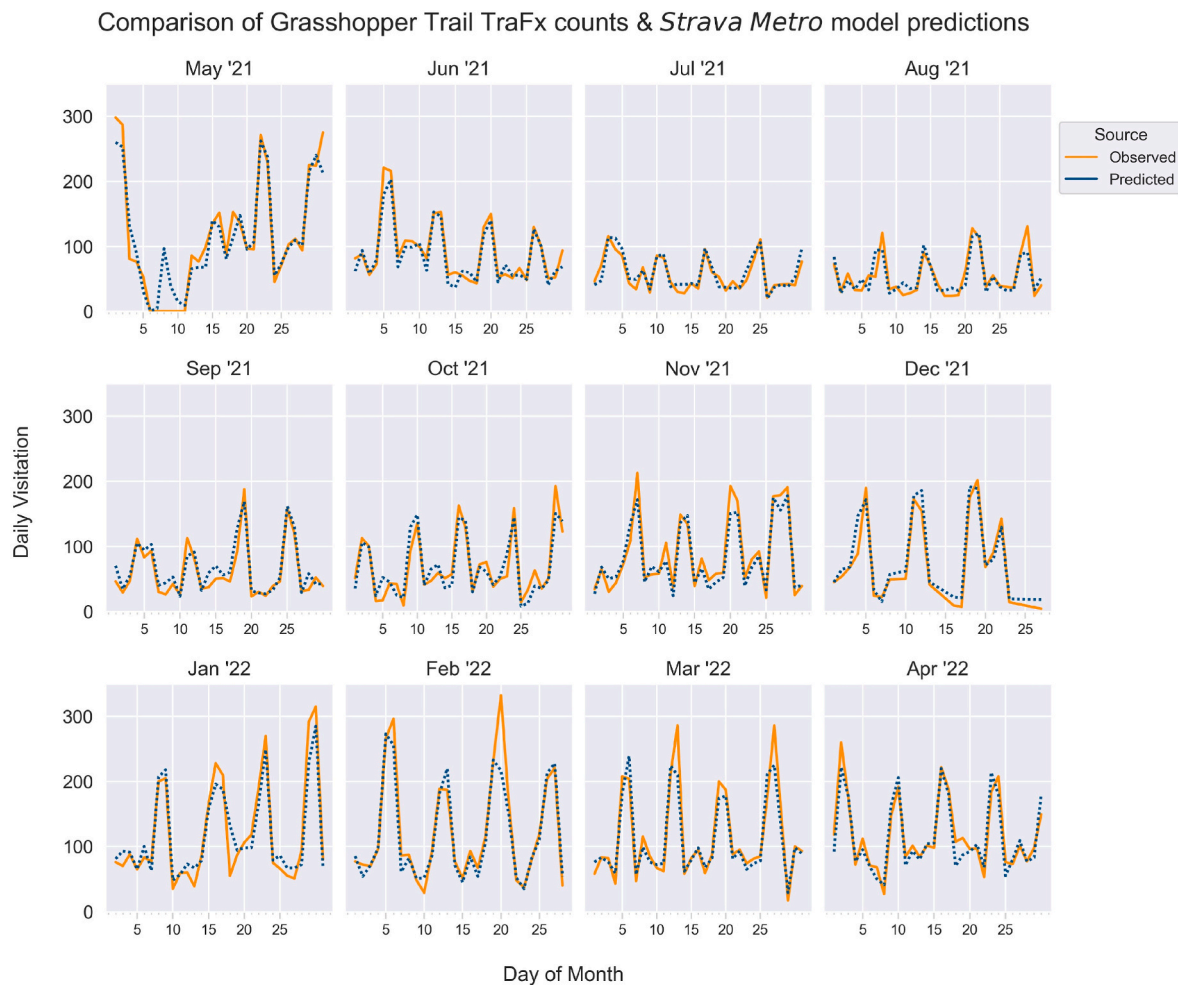


Fig. 4. Daily trail use predictions (blue dotted line) for Grasshopper trail plotted against the TRAFx data (orange solid line) from May 2021 to April 2022. Generally, the peaks of the daily visitation lines correspond to weekend days, while the troughs correspond to weekdays. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4.3. *Strava metro* demographic & activity-type Biases and limitations

Despite the strong correlation with observed trail use and the cost-effectiveness of using *Strava Metro* data to estimate trail use, *Strava* app users contributing to *Strava Metro* were not representative of the population of visitors compared to our visitor-intercept survey sample. However, the visitor intercept survey sample was collected during the summer months (June and August) which we compared to the *Strava Metro* data spanning the calendar year. Although we would expect less seasonal variation in visitor demographics due to the mild, Southern California climate compared to northern latitudes, we cannot rule out the possibility that our survey sample was not representative of the population of visitors to these trails across the calendar year. Additionally, *Strava Metro* combines all pedestrian use-types into a single category, so it's difficult to determine the coverage of *Strava Metro* data for hikers and walkers vs runners, which may have different demographic profiles and different spatial and temporal patterns of use.

Our results were strikingly consistent with [Venter et al. \(2023\)](#), who found *Strava* users to be more likely to be male and cyclists, and a greater proportion of adults age 35–54 than the survey sample. Despite this bias, this cohort of trail users might be a population of interest for managers to plan for safe and satisfactory trail conditions in multiple-use trail settings. However, the gender, activity-type, and age bias of *Strava* users represents a notable limitation of using *Strava Metro* without calibration and model parameterization to account for the demographic biases of the data. As a result, when visitor generated and social media data are used to estimate recreation use, the efficiencies in data collection and monitoring can come at the cost of representativeness of the data. Using *Strava Metro* data in lieu of a statistical sampling approach to estimate visitor demographics would systematically underrepresent female, pedestrian use-types, and older visitors who represent a significant proportion of the constituency that protected areas are intended to serve and provide a diverse range of recreation opportunities and experiences.

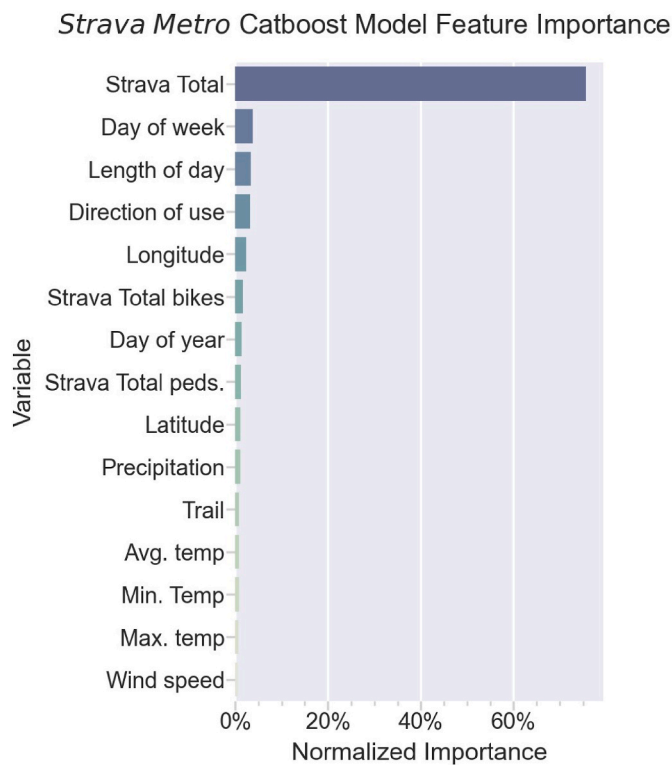


Fig. 5. Normalized variable importance in Catboost model for tree splitting. Strava Total is the most important variable for tree splitting, followed by day of the week, length of day, and direction of use.

5. Conclusion

Visitor generated data from social media and fitness tracking applications provide a rich source of information to better understand and predict recreation use in protected area settings. Numerous studies have demonstrated the efficacy of modelling of social media data for predicting recreation use or visitation at varying spatial and temporal scales, however some of these data serve as an analog (e.g., photo-user-days) for measuring recreation use. Mobile device data sources often obfuscate observed device pings in the interest of protecting individual privacy, or decouple the data from the objective of the measurement, providing estimates based produced by proprietary, black-box models. A

considerable benefit of *Strava Metro* data, relative to social media data or mobile device data streams, is the tangible provenance from a real GNSS/GPS-trace of a specific activity to an aggregated measure of trail use. This direct connection between the data source and objective of the measurement provides a level of transparency, measurement validity, and epistemic coherence for the target of measuring recreation use.

Our findings provide support for the potential of *Strava Metro* data as a tool for monitoring recreation use in protected area settings. We demonstrated that a model incorporating *Strava Metro* data, along with weather and contextual variables, can produce highly accurate predictions of daily trail use, particularly on trails popular with cyclists. This capability offers protected area managers a cost-effective method for understanding fine-scale temporal patterns of visitor use, which can inform operational decisions such as trail maintenance, where to allocate resources for patrol and visitor safety during peak-use periods. The ability to generate reliable daily estimates represents a substantial improvement over many visitor-generated and social media-based data sources, enabling a fine-scale understanding of temporal patterns of recreation use.

However, our research also evaluated and quantified an important limitation: the demographic composition of *Strava* users is not representative of the broader visitor population. Consistent with other studies, we found that *Strava Metro* data is heavily skewed towards male cyclists age 35–54, while underrepresenting pedestrians, females, and older adults which is a notable concern. If managers were to rely solely on this data for visitor profiling, they would risk planning and management decisions based on a narrow segment of visitors, potentially marginalizing a substantial portion of the public. Therefore, while *Strava Metro* is an effective tool for quantifying use, it cannot replace traditional visitor-intercept and statistical sampling methods for characterizing users or understanding more complex aspects of their behavior or recreation experience.

In conclusion, this study advocates for the importance of an integrated approach to recreation monitoring. Protected area managers can leverage the strengths of novel data sources, such as *Strava Metro* data, for continuous, cost-effective, and fine-scale estimation of recreation use, while acknowledging the demographic biases inherent in the data. An effective monitoring strategy would combine these visitor-generated data with periodic, traditional survey methods to maintain an accurate understanding of visitor characteristics and model calibration. Future research could focus on developing robust calibration techniques for modelling visitor use and exploring the transferability approaches across broad geographic and protected area designations, including areas where cycling is limited or prohibited. Additionally, this study illustrates

Table 2

Results of chi-square test of homogeneity comparing visitor demographic characteristics gender, activity-type, and age between Strava Metro and a random intercept survey sample collected in 2021 and 2022.

Variable	Level	Strava Metro	Visitor Intercept Survey	Chi-Square
Gender	Male	40,350 (93.5 %)	667 (73.2 %)	$\chi^2(2, N = 44,056)$ 616.12, $p < .001$
	Female	2345 (5.4 %)	225 (24.7 %)	
	Unspecified	450 (1.0 %)	19 (2.1 %)	
Activity-type	Pedestrian	2850 (6.4 %)	531 (50.8 %)	$\chi^2(1, N = 45,470)$ 2917.59, $p < .001$
	Cyclist	41,575 (93.6 %)	514 (49.2 %)	
Age	18–34	9560 (24.4 %)	200 (23.4 %)	$\chi^2(3, N = 40,106)$ 1128.41, $p < .001$
	35–54	24,685 (62.9 %)	376 (43.9 %)	
	55–64	4870 (12.4 %)	211 (24.6 %)	
	65+	135 (0.3 %)	69 (8.1 %)	

an improvement in the ability to estimate recreation use and human presence and behavior in protected area settings which can be integrated with other data sources for planning and management of protected areas in a manner that is compatible with the habitat conservation goals of PPAs. By thoughtfully combining traditional and novel visitor-generated data sources, managers can assemble a comprehensive and nuanced understanding of recreation use that supports both conservation goals and the provision of high-quality recreation opportunities for all visitors.

CRedit authorship contribution statement

Noah Creany: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Christopher Monz:** Writing – review & editing, Project administration, Methodology, Conceptualization. **Wayne Freimund:** Writing – review & editing, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share the Strava Metro data, but the code for analysis and modelling can be made available upon request from the corresponding author.

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